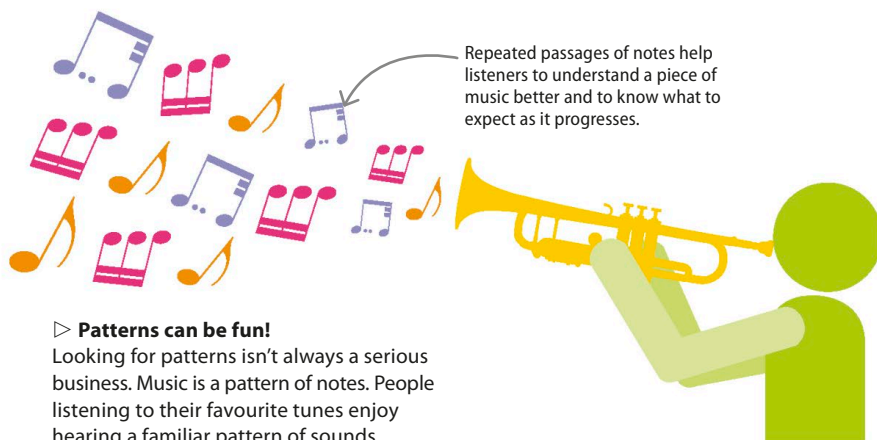


Patterns

Recognizing patterns is an important part of everyday life as it makes interacting with the world easier and more efficient. It's also an important part of computational thinking.

Identifying patterns

A pattern involves repetition in a predictable manner, and lets anyone who recognizes it draw a conclusion. It could be something as simple as a pattern of repeated shapes on a plate. This pattern helps in identifying the plate as part of a particular set of crockery. Recognizing patterns also helps people understand the world more easily. A small vehicle with an engine, four wheels, and no wings is likely to be a car. Its behaviour can also be predicted – it can move at potentially dangerous speeds, but not take off and fly away.



▷ Patterns can be fun!

Looking for patterns isn't always a serious business. Music is a pattern of notes. People listening to their favourite tunes enjoy hearing a familiar pattern of sounds.

SEE ALSO

◀ 68–69 What is computational thinking?

◀ 72–73 Abstraction

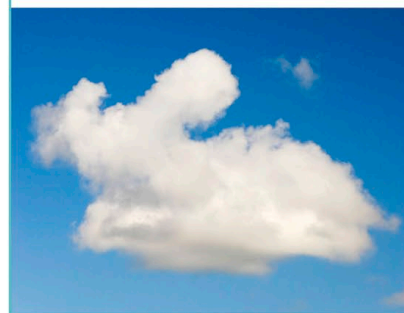
Databases

88–89 ▶

REAL WORLD

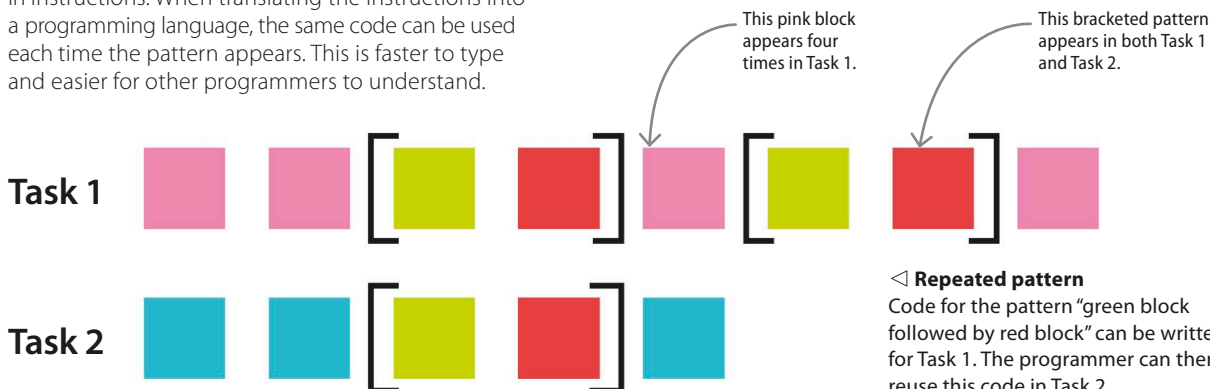
Seeing patterns

The human brain is constantly searching for familiar patterns. So much so that people often see patterns that aren't there. This phenomenon, where a person might think a cloud looks like a particular object, or see a face in a piece of toast, is known as pareidolia. Computers can only recognize faces in pictures if humans write programs that train them what to look for.



Using patterns

Computer scientists look for repeated patterns of steps in instructions. When translating the instructions into a programming language, the same code can be used each time the pattern appears. This is faster to type and easier for other programmers to understand.



◁ Repeated pattern

Code for the pattern "green block followed by red block" can be written for Task 1. The programmer can then reuse this code in Task 2.